3^k Factorial Designs

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Introduction

The 3^k factorial design uses three coded levels per factor — low (-1), middle (0), and high ($+1$) — and runs all 3^k combinations. The third level is what distinguishes the 3^k family from the simpler 2^k family: with three levels the design supports estimation of pure quadratic terms in each factor, and therefore the second-order polynomial approximation to a smooth response surface. The 3^k factorial is the simplest full-factorial construction that captures curvature and is the conceptual ancestor of the modern response-surface designs (central composite, Box-Behnken) that economise on the run count by replacing the dense 3^k grid with sparser alternatives carrying the same second-order information.

Prerequisites

Familiarity with 2^k factorial designs, the concept of an interaction effect, and quadratic regression for capturing curvature in a response.

Theory

For k three-level factors a full factorial has 3^k runs. With coded levels $\{-1, 0, +1\}$ the standard analysis model is the second-order polynomial

$$y = \beta_0 + \sum_i \beta_i x_i + \sum_i \beta_{ii} x_i^2 + \sum_{i<j} \beta_{ij} x_i x_j + \varepsilon,$$

with linear, pure-quadratic, and two-factor-interaction terms. The design grows quickly: $3^3 = 27$, $3^4 = 81$, $3^5 = 243$. For screening purposes the 3^k becomes impractical at $k = 4$ or beyond, and fractional 3^k designs or central composite / Box-Behnken layouts are far more economical for the same second-order information.

Assumptions

The response is well approximated by a second-order polynomial within the design region (no strong cubic or higher curvature), residuals are approximately Normal with constant variance, and the three levels span a range over which the polynomial approximation is plausible.

R Implementation

```
library(rsm)

set.seed(2026)
df <- expand.grid(A = c(-1, 0, 1), B = c(-1, 0, 1))
df$y <- 10 + 2 * df$A + 1.5 * df$B - df$A^2 +
        0.8 * df$A * df$B + rnorm(nrow(df), 0, 0.5)

fit <- rsm(y ~ SO(A, B), data = df)
summary(fit)
```

Output & Results

`rsm()` with the second-order specification returns coefficients for linear, pure-quadratic, and two-factor-interaction terms together with a stationary-point analysis (eigenvalue classification of the quadratic-form matrix as maximum, minimum, or saddle) and a lack-of-fit test against centre replicates when present. Reporting all components of the canonical analysis is the standard summary.

Interpretation

A reporting sentence: "The 3^2 full factorial estimated linear effects of A (2.0, $p < 0.001$) and B (1.5, $p = 0.002$), a significant quadratic term in A (-0.9, $p = 0.004$), and a mild interaction (0.7, $p = 0.07$). The stationary point of the fitted surface lay inside the design region and was a maximum, consistent with the expected concave behaviour. Predicted optimum was 12.4 at $(A, B) = (0.9, 0.8)$." Always pair the coefficient table with the canonical-analysis classification.

Practical Tips

- For $k \geq 3$, prefer central composite or Box-Behnken designs over full 3^k factorials; both deliver the same second-order information with far fewer runs and are the modern industrial-DoE default for response-surface modelling.
- Replicate the centre point ($x = 0$ for all factors) several times in any three-level design; the centre replicates supply pure-error degrees of freedom and a lack-of-fit test that diagnoses inadequacy of the second-order model.
- Coded levels $\{-1, 0, +1\}$ keep the design balanced and orthogonal, ensuring that linear, quadratic, and interaction effects are estimated independently and on directly comparable scales.
- Quadratic effects matter only when the response shows genuine curvature within the design region; if a 2^k factorial with replicated centre points fails to detect curvature, a 3^k or RSM follow-up is unnecessary.
- A 3^k full factorial is often a precursor to formal response-surface methodology; once the second-order surface is mapped, sequential experimentation along the path of steepest ascent or descent locates the optimum.
- For mixed-level designs (some factors at three levels, others at two) use orthogonal-array layouts via `DoE.base`; full crossed factorials become quickly unwieldy in mixed-level cases.

R Packages Used

`rsm` for second-order polynomial fitting with stationary-point and canonical analysis; `FrF2.mix` for fractional three-level and mixed-level designs; `DoE.base` for general orthogonal-array construction including 3^k as a special case; `agricolae` for textbook three-level layouts; `Vdgraph` for variance-dispersion comparisons across 3^k, CCD, and BBD alternatives.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design

- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans